Supplementary C. CRAT Sensitivity Analysis

This supplementary analysis details the sensitivity analysis of CRAT.

C.1 Evaluation design and reporting basis

CRAT's sensitivity on OPUS100 depends on the language pair, evaluation metric, and parameter being varied. Head-selection budget and evaluation top k produce the clearest quality-cost trade-offs, whereas integration rank has a comparatively small effect on BLEU and chrF over the tested range. Regularization changes are often modest, but stronger KL regularization on EN-FA exposes a substantial disagreement between lexical overlap and COMET. Bidirectional averages also conceal several opposing changes between translation directions.

The experiments comprise 85 reported settings for EN-FR, EN-FA and EN-ZH. Seven parameter sweeps contain training measurements, covering 73 settings; the remaining 12 rows vary evaluation top k and have no training measurements. We analyze the latter as an inference-setting sweep, consistent with their fixed parameter counts and unchanged pre-mask mean gates.

Table 1. Tested parameter grids and defaults in the supplied main implementations.

Parameter	Code default	EN-FR values	EN-FA and EN-ZH values
KL coefficient	0.1	0.05, 0.1, 0.2	0.05, 0.1, 0.2, 0.5
Causal coefficient	0.1	0.05, 0.1, 0.2	0.05, 0.1, 0.2, 0.5
Morphology coefficient	0.1	0.05, 0.1, 0.2	0.05, 0.1, 0.2, 0.5
Sparsity coefficient	0.002	0.001, 0.005, 0.01	0.001, 0.005, 0.01, 0.2
KL temperature	2	2, 4, 8	1, 2, 4, 8
Integration rank	16	8, 16, 32	8, 16, 32
Heads per direction	64	32, 64, 96	32, 64, 96
Evaluation top k	2	1, 2, 3, 4	1, 2, 3, 4

EN-FA and EN-ZH use the same grids. The code-default sparsity coefficient, 0.002, is absent from the exports.

BLEU and chrF are equal-weight averages of the forward direction, English to the paired language, and the reverse direction into English. We use the analogous directional macro-average for COMET. Joining the separate COMET exports provides values for all 25 EN-FR settings, all 30 EN-FA settings, and the four EN-ZH top-k settings. All overlapping primary columns agree across the summary, efficiency, routing, and COMET exports, and all 48 supplied BLEU/chrF range summaries reconcile with the primary means.

C.2 Overall sensitivity across parameters

The largest EN-FR ranges occur for top k, at 0.443 BLEU and 0.538 chrF. EN-FA has its largest BLEU range under the KL-coefficient sweep, 0.673, and its largest chrF range under the head-budget sweep, 0.798. EN-ZH is most sensitive in BLEU to top k, 2.148, followed by head budget, 1.558, and the KL coefficient, 1.506. Its largest full-grid chrF range, 0.667, occurs under the KL coefficient. These ranges describe the tested grid; they are not uncertainty intervals or evidence of statistical significance.

The larger EN-ZH BLEU sensitivity persists on matched grids. In contrast, its KL-coefficient chrF range falls from 0.667 to 0.169 when the extra value 0.5 is excluded, showing that extreme settings can substantially change a range-based ranking. Integration rank remains comparatively insensitive across all pairs, with BLEU ranges of 0.072-0.087 and chrF ranges of 0.001-0.082.

Table 2. Maximum minus minimum mean quality over each complete reported grid.

Parameter	EN-FR BLEU	EN-FR chrF	EN-FA BLEU	EN-FA chrF	EN-ZH BLEU	EN-ZH chrF
KL coefficient	0.195	0.255	0.673	0.506	1.506	0.667
Causal coefficient	0.062	0.080	0.233	0.599	0.182	0.056
Morphology coefficient	0.236	0.286	0.106	0.188	0.127	0.152
Sparsity coefficient	0.084	0.109	0.221	0.081	0.160	0.269
KL temperature	0.186	0.149	0.530	0.567	0.080	0.189
Integration rank	0.074	0.082	0.087	0.001	0.072	0.040
Heads per direction	0.270	0.393	0.627	0.798	1.558	0.652
Evaluation top k	0.443	0.538	0.100	0.145	2.148	0.144

A smaller range means less variation in the reported means on that grid. Grid widths and counts differ across pairs for several parameters. Figure 1 therefore repeats the comparison using only parameter values shared by all three pairs.

C.3 Sensitivity to the training objectives

KL coefficient. On EN-FR, 0.05 gives the highest BLEU, chrF, and COMET within this sweep: 31.654, 55.808, and 0.80672. Relative to the sweep's 0.1 row, the increases are 0.195 BLEU, 0.255 chrF, and 0.00063 COMET. Increasing the coefficient to 0.2 also improves all three means relative to 0.1, but less than 0.05. The nonmonotonic response does not support a simple rule that more regularization always helps.

On EN-FA, increasing the KL coefficient from 0.1 to 0.5 raises mean BLEU from 8.455 to 8.870 and chrF from 26.007 to 26.172, while reducing COMET from 0.65678 to 0.64422. The COMET decrease of 0.01255 is the largest range among the training-objective sweeps for this pair. The directional scores clarify the conflict: English-to-Persian BLEU decreases by 0.418 and COMET by 0.02600, whereas Persian-to-English BLEU increases by 1.248. Thus, the improved average BLEU does not indicate an improvement in both directions. The validation loss also increases from 1.26294 to 1.28626, giving no support for selecting 0.5 solely because it maximizes the reported mean BLEU.

On EN-ZH, 0.2 gives the highest BLEU, 25.871, an increase of 1.506 over the sweep's 0.1 row, with a small chrF increase of 0.019. At 0.5, BLEU remains 1.267 above the 0.1 reference but chrF falls by 0.498. The highest chrF instead occurs at 0.05. These results favor different choices depending on the metric.

Causal-prior coefficient. EN-FR is comparatively insensitive over 0.05-0.2: the ranges are 0.062 BLEU, 0.080 chrF, and 0.00046 COMET, with 0.1 highest on all three. EN-FA benefits in mean lexical scores from increasing the coefficient to 0.5, reaching 8.707 BLEU and 26.520 chrF, gains of 0.170 and 0.532 over 0.1. COMET instead peaks at 0.2, although its advantage over 0.1 is only 0.00016. EN-ZH shows a 0.182 BLEU range and a 0.056 chrF range; 0.1 has the highest BLEU and 0.5 the highest chrF. The small differences in several of these comparisons should be interpreted as observed rankings rather than established improvements.

The causal coefficient regulates alignment to the direction-conditioned priors in the earlier implementation. Its effects should therefore be described as sensitivity to the prior-alignment objective, without equating a coefficient change to a new causal discovery procedure. The routing results in Section 7 show that the direction of the gate response also varies by language pair.

Morphology coefficient. EN-FR favors 0.2 within the tested range, with increases of 0.236 BLEU, 0.275 chrF, and 0.00051 COMET over the 0.1 row. EN-FA is more mixed: 0.1 and 0.2 differ by only 0.000014 BLEU, 0.05 gives the highest chrF, and 0.1 gives the highest COMET. The near-tie in BLEU cannot justify a substantive preference for 0.2. On EN-ZH, 0.05 has the highest BLEU and 0.2 the highest chrF, with total ranges of 0.127 and 0.152. These results support a pair-dependent response to the auxiliary morphology objective, not a universal coefficient or a conclusion about morphological accuracy from aggregate translation scores alone.

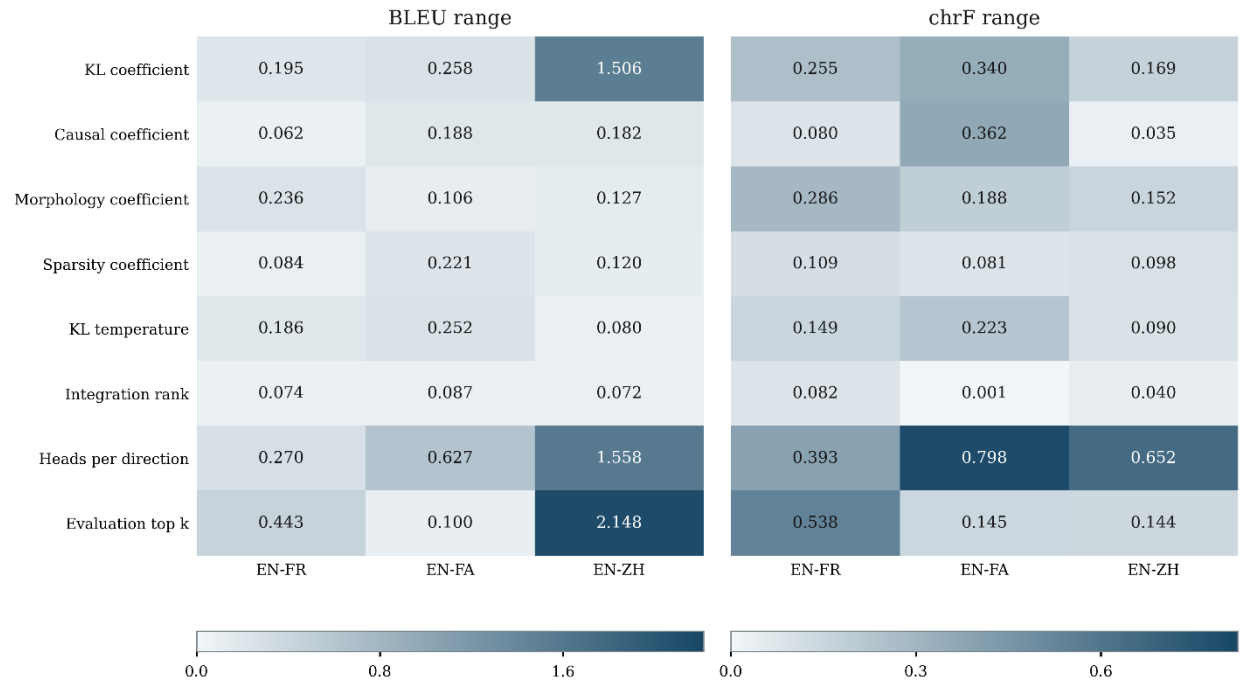


Figure 1. Ranges of reported BLEU and chrF means on parameter grids shared by all three pairs. Shared values are 0.05/0.1/0.2 for KL, causal, and morphology coefficients; 0.001/0.005/0.01 for sparsity; 2/4/8 for temperature; 8/16/32 for rank; 32/64/96 for head budget; and 1/2/3/4 for top k. Each metric has its own color scale. Ranges describe sensitivity of means, not statistical uncertainty.

Sparsity coefficient. The original implementation's default, 0.002, is absent from every supplied sweep. We therefore compare the observed settings directly and do not treat the exported differences from an unseen default as verified effect sizes. EN-FR improves modestly from 0.001 to 0.01, with increases of 0.084 BLEU, 0.109 chrF, and approximately 0.00003 COMET between these endpoints. Its COMET minimum occurs at 0.005, so the full COMET range is larger, 0.00057. EN-FA has different best settings for each metric: 0.001 for BLEU, 0.005 for chrF, and 0.2 for COMET. Its chrF range is only 0.081, despite clear changes in the routing gates. EN-ZH has its highest BLEU at 0.005 and chrF at 0.01; increasing the coefficient to 0.2 lowers chrF to 43.973, compared with 44.242 at 0.01.

The sparsity sweep illustrates why routing behavior and translation quality should be evaluated separately. Stronger regularization consistently reduces mean soft gates, but the lexical response can remain small or nonmonotonic. A lower soft-gate count alone does not demonstrate a quality improvement, fewer selected top-k roles, or faster generation.

KL temperature. This parameter controls the distillation objective, not the decoding temperature. EN-FR has its highest BLEU, chrF, and COMET at temperature 4, improving over temperature 2 by 0.186, 0.139, and 0.00049, respectively. EN-FA has its highest BLEU and chrF at temperature 1, improving by 0.530 and 0.567 over temperature 2, but COMET decreases by 0.00126. The lexical gains are concentrated in the reverse direction: Persian-to-English BLEU rises by 0.935, compared with 0.124 in English-to-Persian, and English-to-Persian chrF decreases slightly. EN-FA COMET peaks at temperature 4. EN-ZH has only a 0.080 BLEU range across temperatures 1-8, with the highest BLEU at 2 and highest chrF at 8. Temperature sensitivity is consequently smaller for EN-ZH BLEU than for EN-FA BLEU over the supplied grids.

The objective-weight and temperature sweeps leave the reported trainable parameter count and peak memory unchanged within each pair. Their timing differences therefore cannot be described as parameter savings. Training times vary across nominally identical defaults as well as across changed coefficients, so small timing differences require repeated measurements before being attributed to the hyperparameter.

C.4 Structural capacity and resource cost

On EN-FR, the 32-head budget gives the highest mean BLEU, chrF, and COMET among the three tested budgets, while using the fewest parameters and the least peak memory and training time. Relative to 64 heads, it improves quality by 0.270 BLEU, 0.393 chrF, and 0.00262 COMET; reduces trainable parameters by 48.56%; reduces peak memory by 7.18%; and reduces training time by 11.30%. It also increases batched generation throughput by 8.40%. This is an observed improvement across quality and measured resource costs within the head-budget sweep, subject to the reported variability.

EN-FA benefits in all three aggregate quality means from the largest budget, 96 heads, which improves over 64 heads by 0.627 BLEU, 0.792 chrF, and 0.00493 COMET. The improvement carries a substantial cost: 41.85% more trainable parameters, 5.84% higher peak memory, 14.40% longer training, and 18.27% lower batched throughput. Mean single-example latency increases from 0.832 s to 1.601 s, and the 95th percentile from 3.137 s to 6.308 s. The 32-head alternative saves 44.29% of trainable parameters but lowers COMET by 0.00508, despite a 0.469 BLEU increase.

EN-ZH presents a direct quality trade-off. Moving from 64 to 32 heads increases BLEU by 1.234 and reduces parameters by 46.36%, peak memory by 8.51%, and training time by 9.35%, while lowering chrF by 0.599. The BLEU increase comes primarily from English-to-Chinese, where it rises by 2.446. The 96-head budget has the highest aggregate chrF but the lowest aggregate BLEU. Without COMET for this sweep, the evidence does not establish a single quality-optimal budget for EN-ZH.

Integration rank. Increasing rank from 8 to 32 adds exactly 589,824 trainable parameters in each pair. The accompanying peak-memory increase is about 0.0094 GiB, much smaller than the change induced by the head-budget sweep. Rank 32 has the highest BLEU and chrF in all three pairs, but its gains over rank 16 are small: 0.074/0.045 BLEU/chrF on EN-FR, 0.087/0.001 on EN-FA, and 0.060/0.024 on EN-ZH. COMET favors rank 16 on EN-FR and EN-FA. On EN-FA, increasing rank from 16 to 32 decreases COMET by 0.00257 even though chrF is almost unchanged. Thus, the small lexical ranges do not establish insensitivity on every metric.

Table 3. Quality and resource cost under different head-selection budgets.

Heads per direction	Trainable (M)	BLEU	chrF	COMET	Peak (GiB)	Train (min)	Generation tokens/s
EN-FR							
32	102.206	31.736	56.082	0.80889	36.941	19.44	85.31
64	198.675	31.466	55.688	0.80627	39.800	21.92	78.70
96	286.756	31.674	55.784	0.80687	42.094	22.39	76.31
EN-FA							
32	120.032	8.699	25.930	0.64696	37.249	17.13	101.36
64	215.452	8.231	25.936	0.65205	40.116	18.43	98.57
96	305.630	8.858	26.728	0.65698	42.460	21.08	80.57
EN-ZH							
32	127.372	25.759	43.842	0.69409	37.579	18.67	103.93
64	237.473	24.525	44.441	0.70901	41.073	20.60	95.92
96	343.379	24.200	44.493	0.70647	43.678	22.79	92.30

M denotes millions of trainable parameters. Resource entries are reported means. Peak measurements in this table include the training-containing workflow.

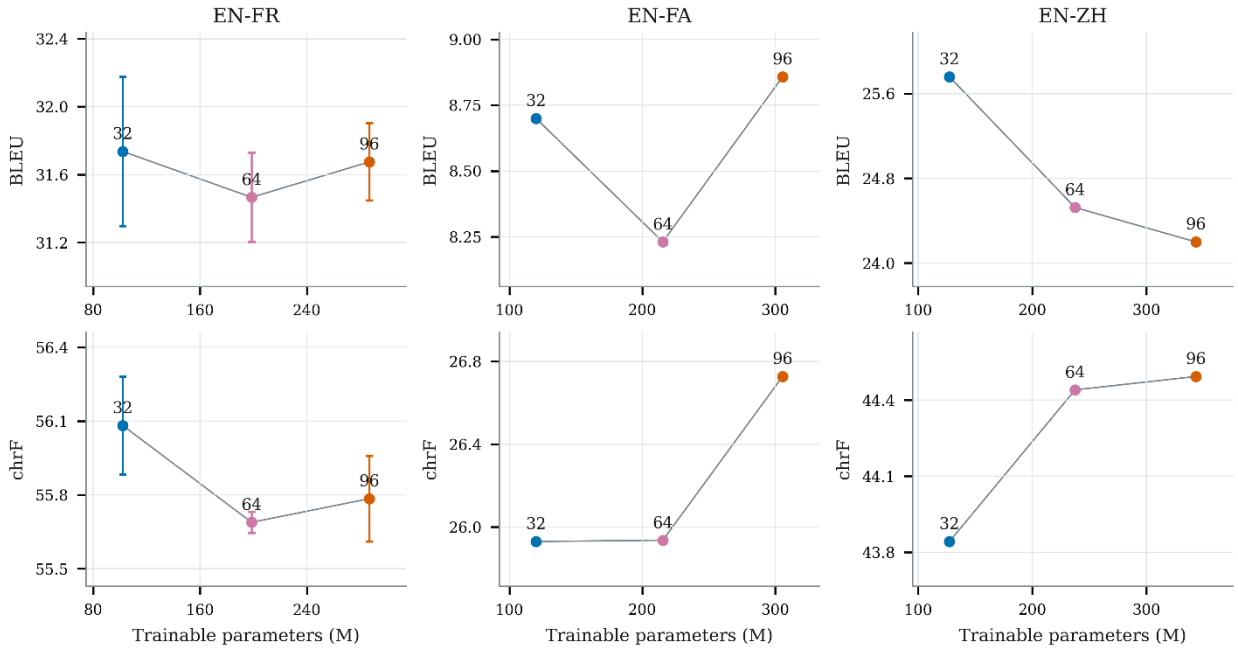


Figure 2. BLEU and chrF versus trainable parameter count under head-budget variation. Labels identify heads per direction. EN-FR bars show ±1 reported SD; no variability estimates are available for EN-FA or EN-ZH. Axis ranges vary by panel. Lines connect ordered tested budgets and do not imply that intermediate settings were evaluated.

C.5 Evaluation top *k*

For EN-FR, top *k* equal to 1 has the highest BLEU, chrF, and COMET, improving over 2 by 0.394, 0.510, and 0.00147. Both translation directions improve on all three metrics. Increasing to 3 or 4 does not improve BLEU or chrF over 2, although their COMET means are slightly higher. The observed quality preference is therefore for a single selected role in this sweep, rather than for the largest number of retained roles.

EN-FA has a different response. Top *k* equal to 4 gives the highest BLEU and COMET, improving over 2 by 0.100 BLEU and 0.01188 COMET, and also improves chrF by 0.049. Top *k* equal to 1 gives the highest mean chrF but reduces COMET by 0.00910. Its English-to-Persian scores deteriorate while reverse-direction lexical scores improve. The COMET range across top-k settings is 0.02098, larger than the COMET range of any other parameter sweep for EN-FA.

EN-ZH favors top *k* equal to 1 for BLEU, giving 26.390 compared with 24.532 at 2. However, 2 has the highest chrF and COMET. Switching from 2 to 1 increases BLEU by 1.858 but decreases chrF by 0.144 and COMET by 0.00159. The result is a metric-specific trade-off, not an unqualified advantage for either choice.

Generation speed does not decrease monotonically as more roles are retained. Top *k* equal to 4 has the highest batched token throughput in all three pairs, exceeding 2 by 12.04%, 5.76%, and 14.70% for EN-FR, EN-FA, and EN-ZH. It also has the lowest mean single-example latency in each pair. Tail behavior differs: EN-FA's 95th-percentile latency is 7.65% higher at 4 than at 2, despite the 10.16% reduction in mean latency. Consequently, neither selected-role count nor mean latency is sufficient to characterize measured inference efficiency.

The reported top-*k* peaks are nearly constant at 15.369, 15.602, and 15.782 GiB within the three pairs. These evaluation-only measurements must not be compared with the approximately 37-44 GiB peaks of training-containing sweeps as if the difference were caused by top-*k* sparsity. The earlier generation timer also excludes tokenization, feature construction, and router computation. Output lengths vary between settings, and the reported throughput and latency remain specific to the measured workload and execution path.

Table 4. Quality and inference measurements across evaluation top-*k* settings.

Top *k*	BLEU	chrF	COMET	Generation tokens/s	Mean latency (s)	p95 latency (s)
EN-FR						
1	31.964	56.094	0.80745	98.80	0.818	2.208
2	31.570	55.584	0.80598	92.57	0.876	2.467
3	31.522	55.556	0.80663	92.41	0.855	2.450
4	31.556	55.572	0.80646	103.72	0.750	2.058
EN-FA						
1	8.461	26.115	0.64895	95.66	0.779	2.658
2	8.399	25.970	0.65805	96.40	0.839	2.968
3	8.428	25.983	0.65715	96.75	0.846	3.183
4	8.498	26.019	0.66993	101.95	0.754	3.195
EN-ZH						
1	26.390	44.010	0.80552	101.59	1.147	2.684
2	24.532	44.155	0.80711	92.37	1.219	3.232
3	24.242	44.037	0.80550	92.75	1.254	3.342
4	24.804	44.032	0.80552	105.94	1.084	2.786

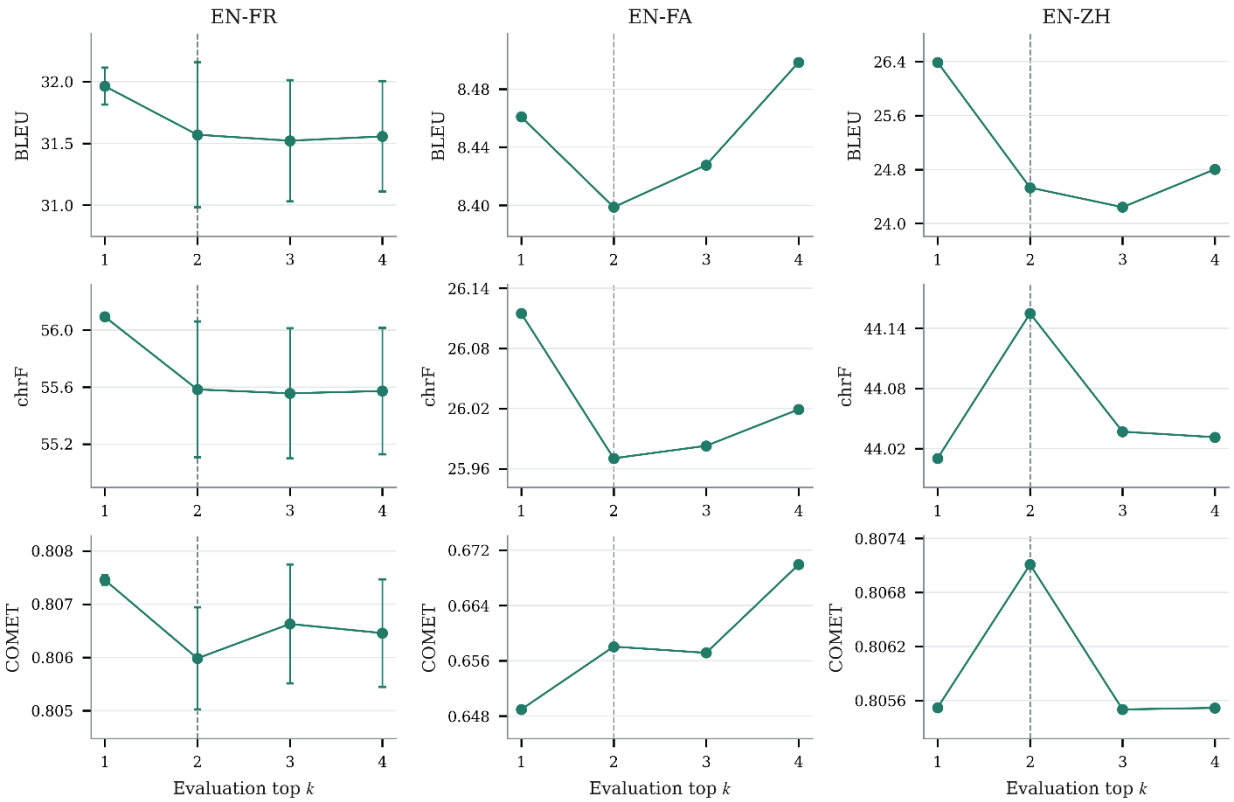


Figure 3. Quality profiles across evaluation top-k settings. The dashed vertical line marks top k equal to 2. EN-FR bars show ±1 reported SD; absence of bars for EN-FA and EN-ZH reflects unavailable variability. Panel scales differ, and error bars are not paired confidence intervals for differences.

C.6 Directional effects and output failures

The directional results explain several apparent improvements in bidirectional averages. With EN-FA top *k* equal to 1, English-to-Persian BLEU decreases by 0.507 and COMET by 0.01884 relative to top k equal to 2. Persian-to-English BLEU increases by 0.631, producing a positive aggregate BLEU change even though the forward direction worsens. At top *k* equal to 4, the pattern reverses: English-to-Persian BLEU increases by 0.365 and COMET by 0.02444, while reverse-direction BLEU decreases by 0.165 and COMET by 0.00067. The resulting mean COMET increase therefore comes from a substantial forward-direction improvement.

For EN-ZH, top *k* equal to 1 increases English-to-Chinese BLEU by 3.131 and Chinese-to-English BLEU by 0.586 relative to 2. English-to-Chinese chrF and COMET nevertheless decrease by 0.631 and 0.00504, while reverse-direction chrF and COMET increase. Thus, even the large BLEU gain is not uniformly corroborated by the other metrics in the forward direction. EN-FR provides a more consistent example: top *k* equal to 1 improves both directions across BLEU, chrF, and COMET.

Table 5. Selected directional differences relative to the corresponding within-sweep reference.

Pair	Contrast	BLEU forward	BLEU reverse	COMET forward	COMET reverse
EN-FA	KL 0.5 vs 0.1	-0.418	+1.248	-0.02600	+0.00089
EN-FA	Temperature 1 vs 2	+0.124	+0.935	-0.00458	+0.00205
EN-FA	Heads 96 vs 64	+0.167	+1.087	+0.00515	+0.00472
EN-ZH	Heads 32 vs 64	+2.446	+0.022	+0.00401	+0.00521
EN-FA	Top k 1 vs 2	-0.507	+0.631	-0.01884	+0.00065
EN-FA	Top k 4 vs 2	+0.365	-0.165	+0.02444	-0.00067
EN-ZH	Top k 1 vs 2	+3.131	+0.586	-0.00504	+0.00187

Forward means English to the paired language; reverse means the paired language to English. Positive differences indicate higher scores. All directional scores and differences are retained in the companion CSVs.

C.7 Routing responses

Stronger sparsity regularization reduces all four evaluation mean soft gates in every pair. At coefficient 0.2, EN-FA's source, indicator, position, and integration means are 0.650, 0.235, 0.726, and 0.496, compared with 0.797, 0.431, 0.847, and 0.652 at 0.001. The mean count above 0.5 falls from 3 to 2. EN-ZH similarly falls from 2.5 to 2, with all

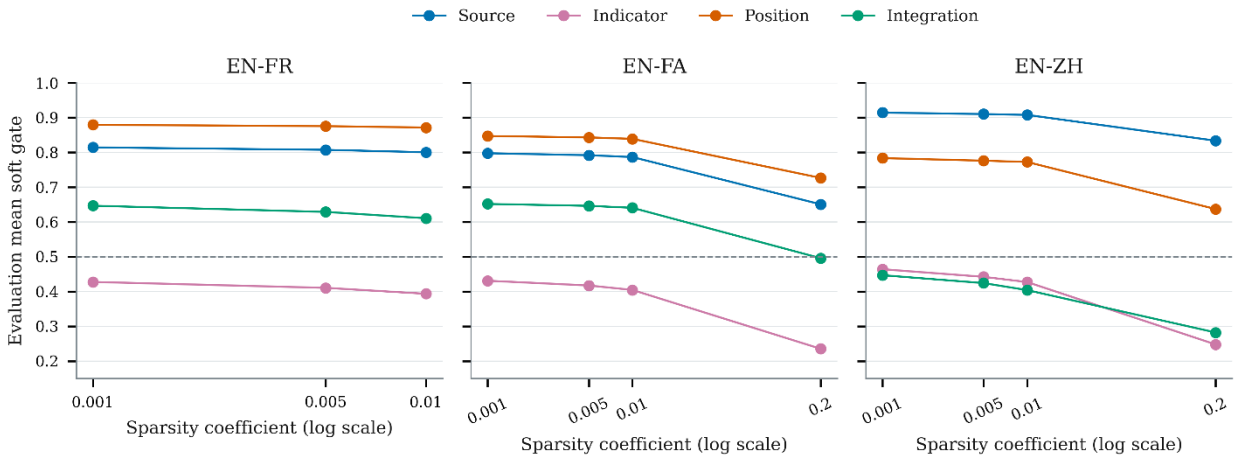


Figure 4. Evaluation mean soft gates across sparsity coefficients. The horizontal axis is logarithmic; EN-FR lacks the 0.2 setting. The dashed line marks the diagnostic threshold of 0.5. These gates precede top-k masking and are not categorical role probabilities.

four mean gates decreasing. EN-FR remains at a count of 3 throughout its narrower 0.001-0.01 sweep. These counts refer to soft gates before top-*k* masking.

The causal coefficient produces different gate responses. Increasing it raises all four evaluation mean gates on EN-FR over 0.05-0.2. On EN-FA over 0.05-0.5, the integration mean increases from 0.647 to 0.659 while the indicator mean decreases from 0.454 to 0.387. EN-ZH also increases integration and decreases indicator, with its soft-threshold count dropping from 2.5 to 2 at 0.5. This variation is consistent with alignment to pair-dependent priors, rather than a universal tendency to increase or decrease all gates.

The explicit selection-rate fields add information beyond mean gates. For EN-FR and EN-FA, the exported rates are 0 or 1: position is selected at 1; source joins at 2; integration joins at 3; and indicator joins at 4. At the reported precision, these sweeps show a fixed selected-role combination for each top-*k* value. They do not demonstrate diverse input-dependent hard role assignments, although the gate magnitudes may still vary across examples.

Table 6. Selected roles and retained soft-gate mass for EN-FR and EN-FA.

Top *k*	Selected roles	Retained gate mass (%)	Soft count above 0.5	Logged gate statistic
EN-FR				
1	Position	32.16	3.0	0.94548
2	Position and source	61.82	3.0	0.94548
3	Position, source, integration	84.92	3.0	0.94548
4	All four roles	100.00	3.0	0.94548
EN-FA				
1	Position	31.26	3.0	0.96282
2	Position and source	60.62	3.0	0.96282
3	Position, source, integration	84.52	3.0	0.96282
4	All four roles	100.00	3.0	0.96282

Selected-role rates are exactly 0 or 1 at the exported precision, and their sum equals top k. Retained mass refers to soft-gate magnitudes, not categorical probabilities or fractions of model FLOPs. The last column preserves top *k* router entropy without asserting categorical entropy. Equivalent top-k selection and mass fields are absent for EN-ZH.